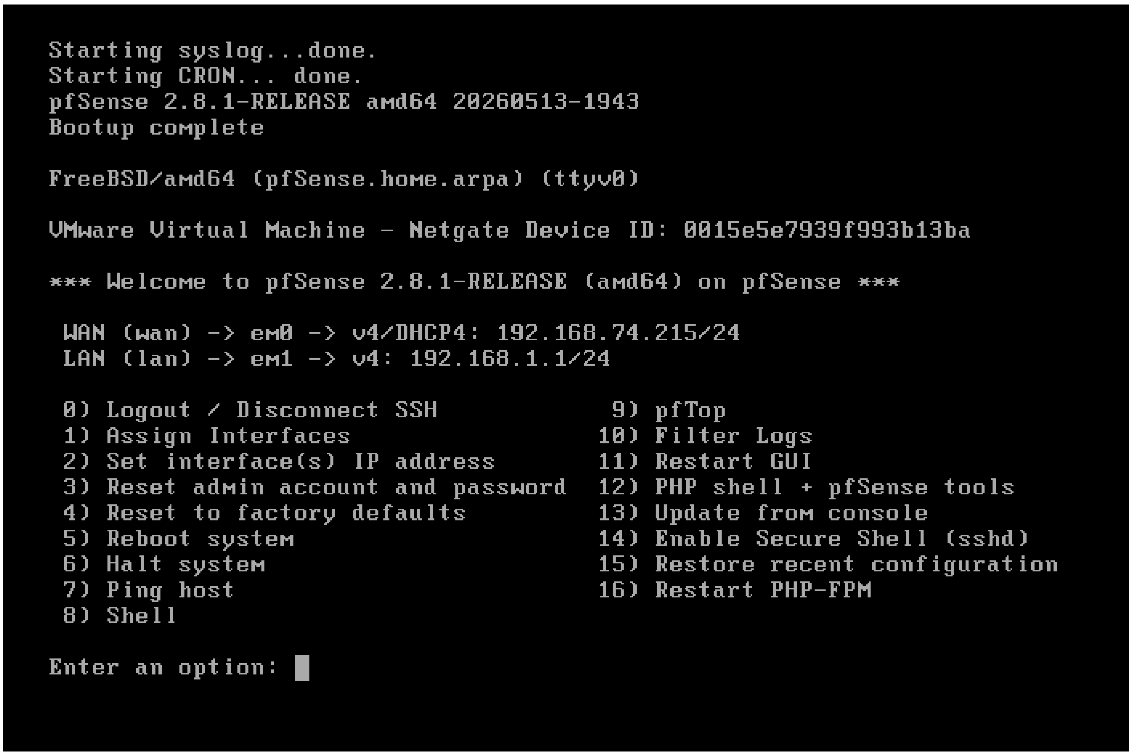
Assignment 2

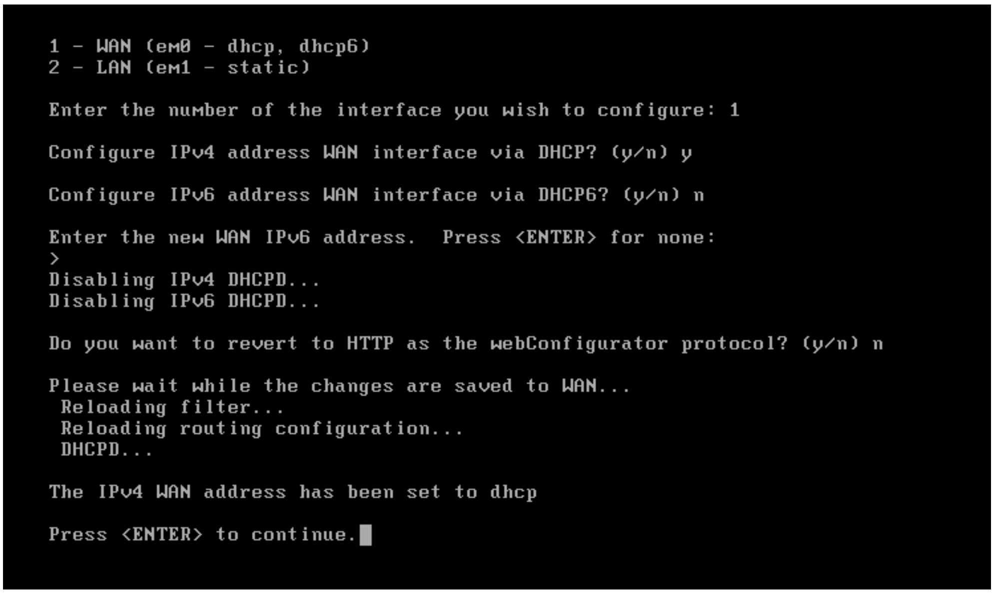
Install and configure pfSense Firewall, Squid, and SquidGuard.

1) **Boot pfSense firewall successfully.**



```
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VMware Virtual Machine - Netgate Device ID: 0015e5e7939f993b13ba

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 192.168.74.215/24
 LAN (lan) -> em1 -> v4: 192.168.1.1/24

 0) Logout / Disconnect SSH            9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart GUI
 3) Reset admin account and password  12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```

2) **Configure WAN interface to obtain IPv4 address using DHCP.**



```
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 1

Configure IPv4 address WAN interface via DHCP? (y/n) y

Configure IPv6 address WAN interface via DHCP6? (y/n) n

Enter the new WAN IPv6 address.  Press <ENTER> for none:
>
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to WAN...
 Reloading filter...
 Reloading routing configuration...
 DHCPD...

The IPv4 WAN address has been set to dhcp

Press <ENTER> to continue.
```

3)Select LAN interface for IP configuration.

```
VMware Virtual Machine - Netgate Device ID: 0015e5e7939f993b13ba

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 192.168.74.215/24
 LAN (lan) -> em1 -> v4: 192.168.1.1/24

 0) Logout / Disconnect SSH            9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart GUI
 3) Reset admin account and password  12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure:
```

4) Assign static LAN IP address 192.168.25.5/24.

```
Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address.  Press <ENTER> for none:
> 192.168.25.5

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
```

5) Disable DHCP server and IPv6 on LAN interface.

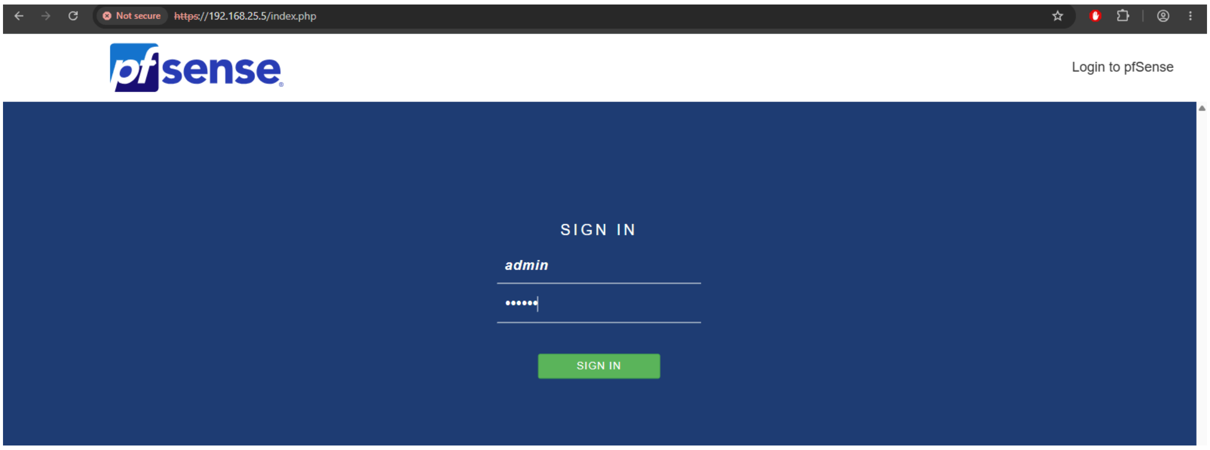
```
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address.  Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to LAN...
 Reloading filter...
 Reloading routing configuration...
 DHCPD...

The IPv4 LAN address has been set to 192.168.25.5/24
You can now access the webConfigurator by opening the following URL in your web
browser:
                https://192.168.25.5/

Press <ENTER> to continue.
```

6) Verify LAN IP configuration and access URL (https://192.168.25.5).

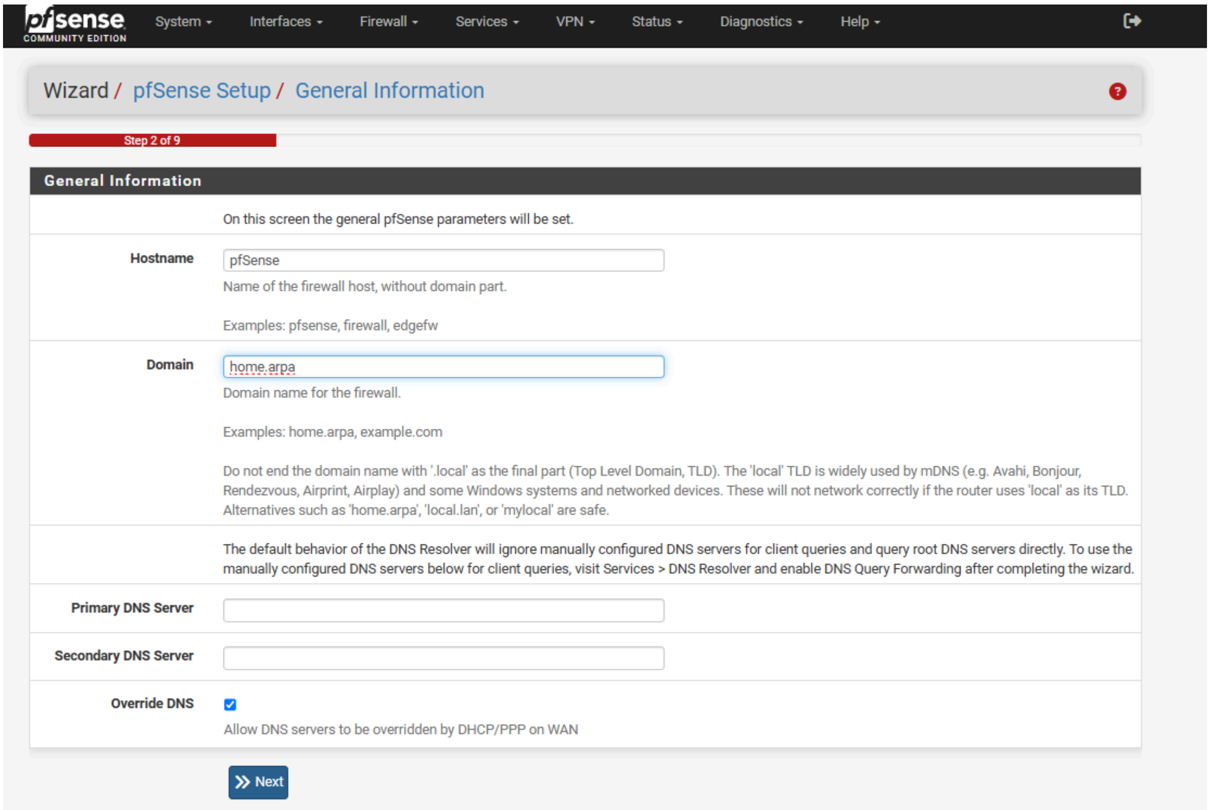
```
The IPv4 LAN address has been set to 192.168.25.5/24
You can now access the webConfigurator by opening the following URL in your web
browser:
                https://192.168.25.5/

Press <ENTER> to continue.
VMware Virtual Machine - Netgate Device ID: 0015e5e7939f993b13ba

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 192.168.74.215/24
 LAN (lan) -> em1 -> v4: 192.168.25.5/24

 0) Logout / Disconnect SSH            9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart GUI
 3) Reset admin account and password  12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option:
```

7) Reset pfSense admin account and set new password.

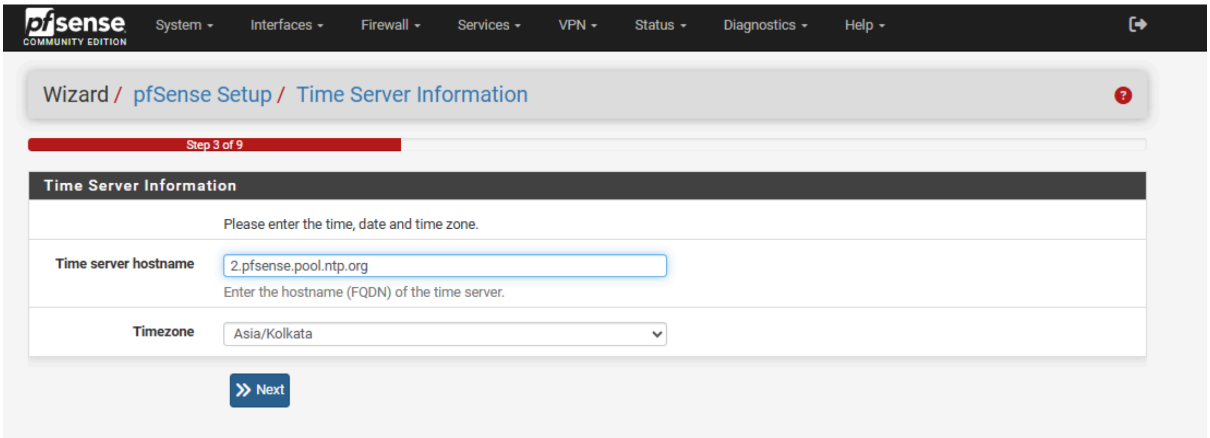
8) Login to pfSense WebConfigurator using admin credentials.



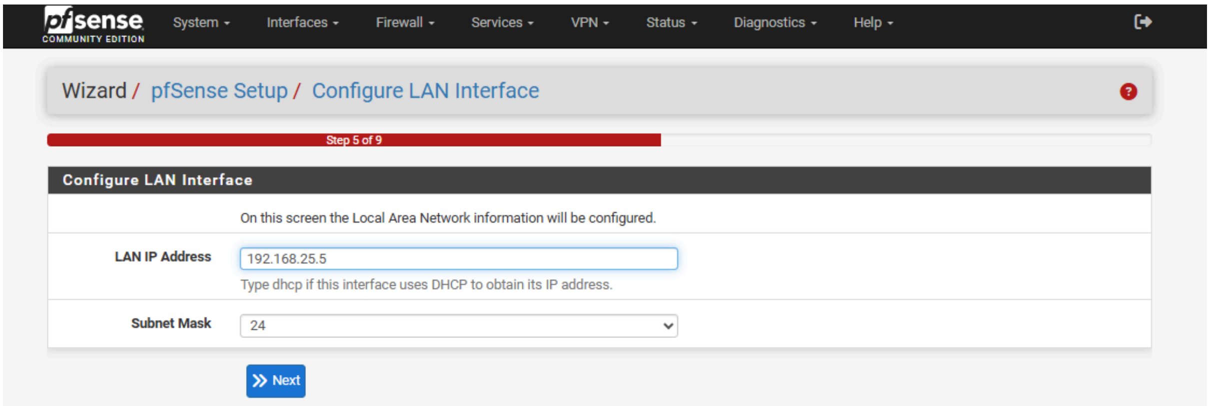
9) Run initial setup wizard and configure hostname/domain.



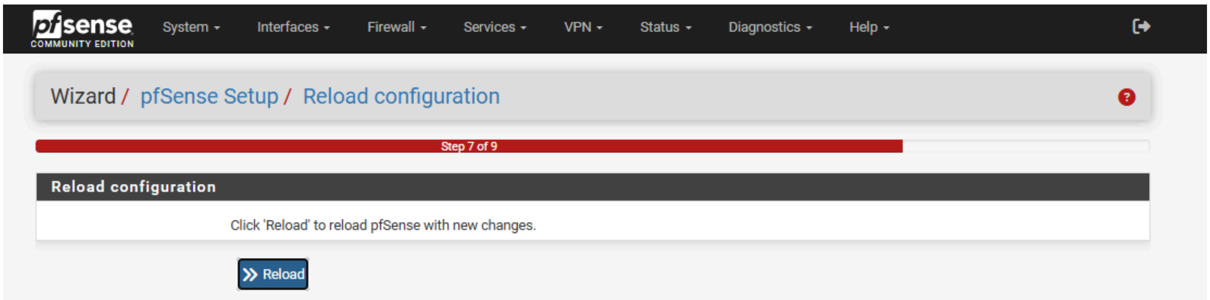
10) Configure NTP time server and select timezone.



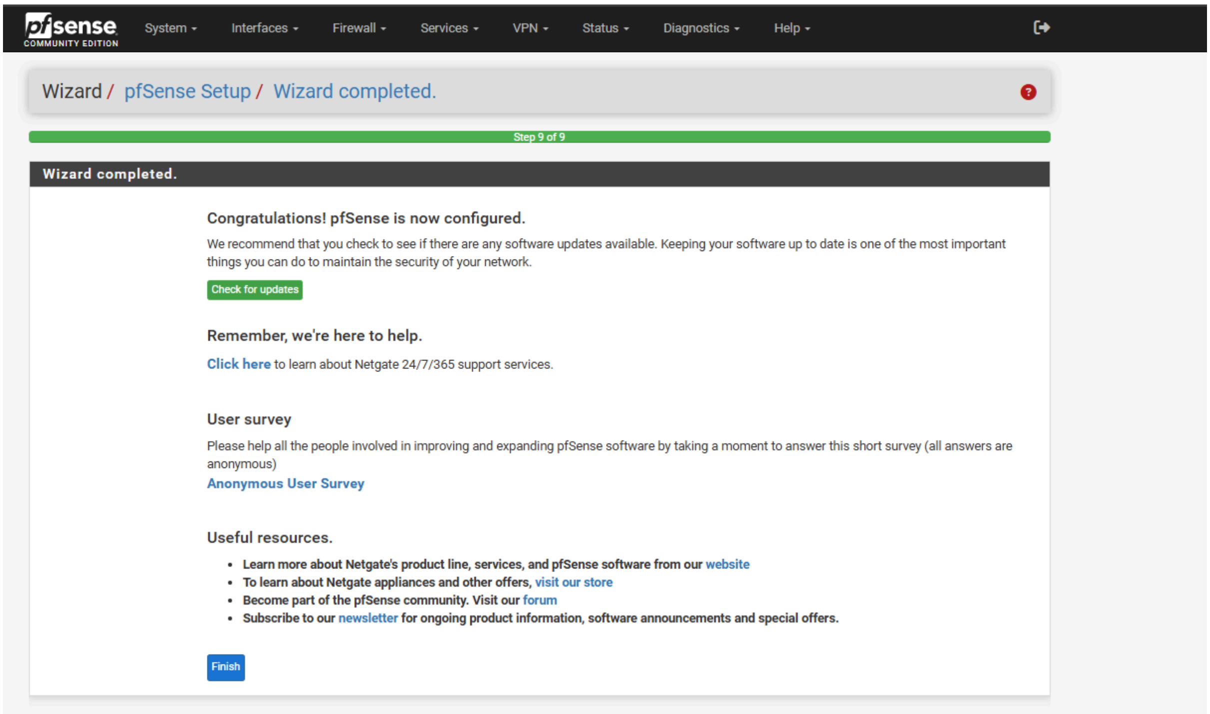
11) Confirm LAN IP address and subnet mask settings.



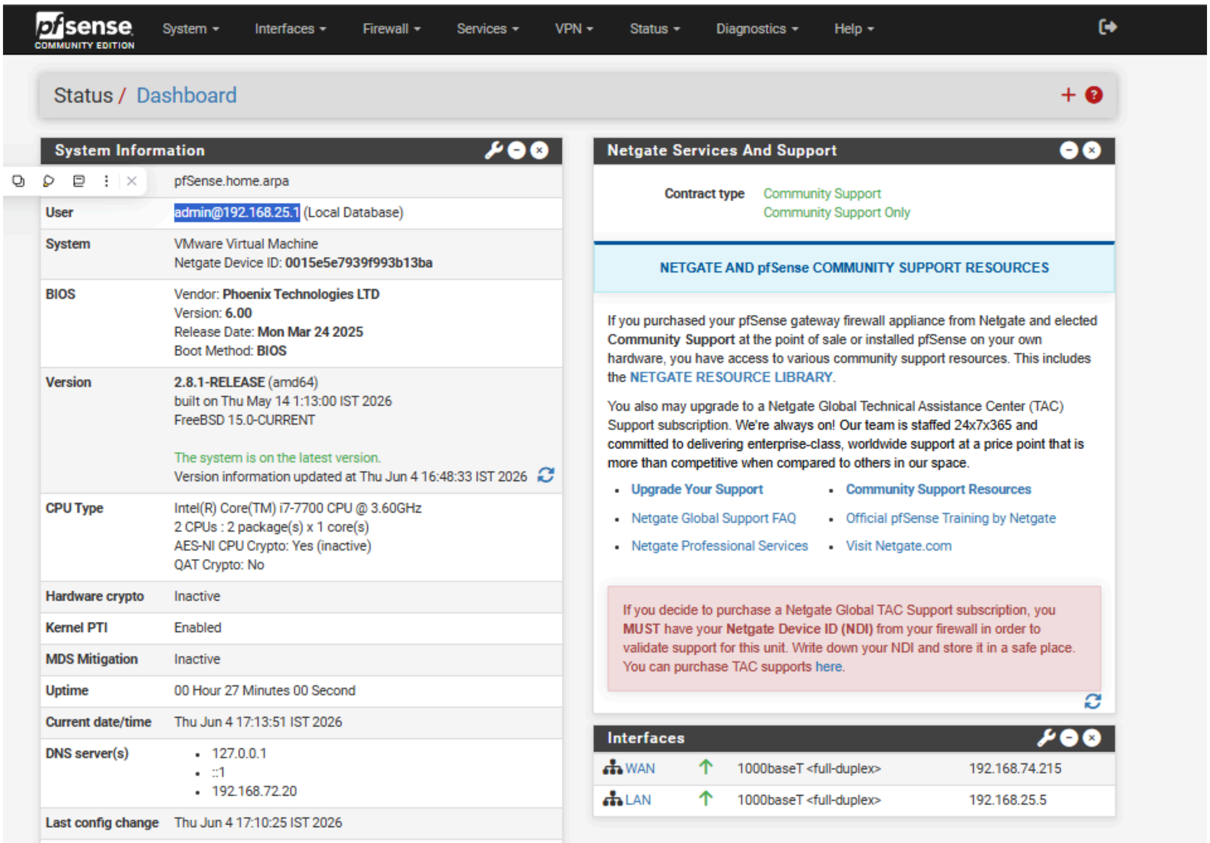
12) Reload pfSense configuration and apply changes.



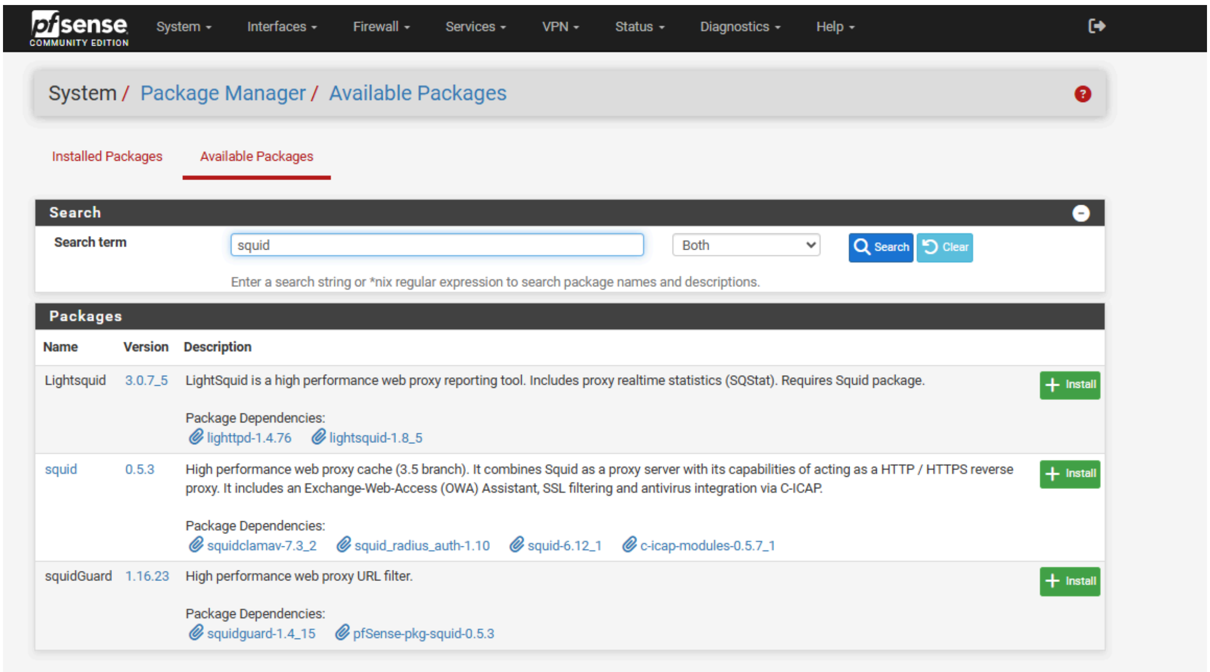
13) Complete setup wizard and verify successful installation.



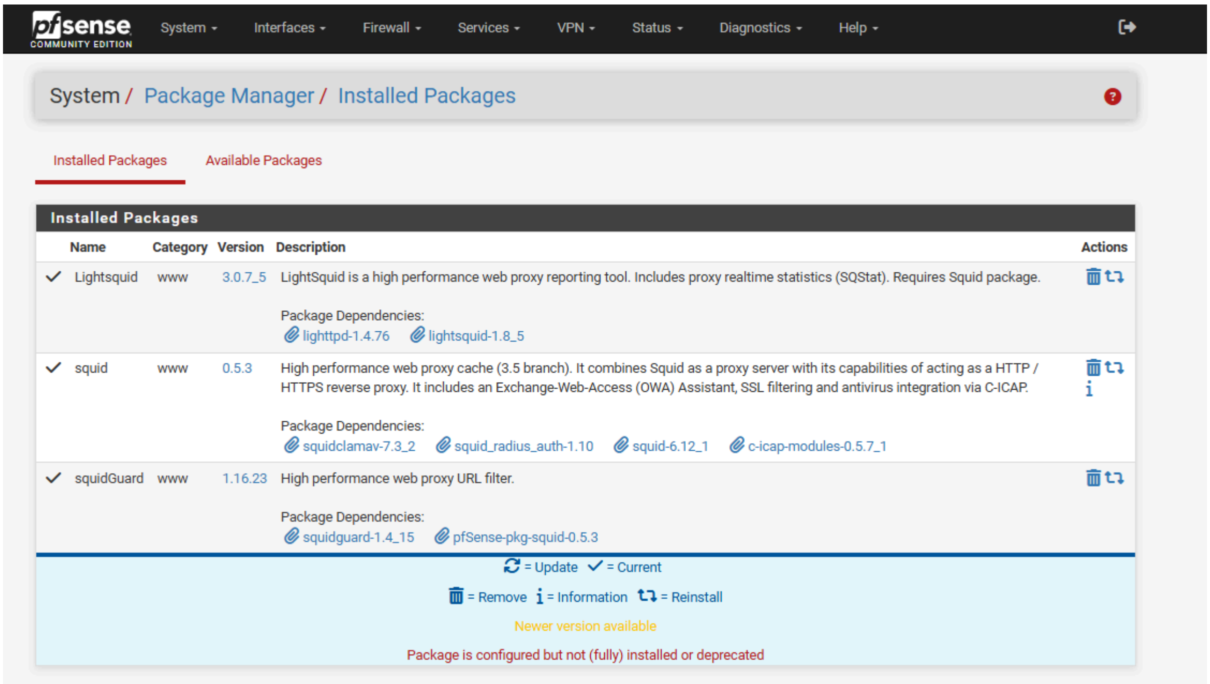
14) Verify pfSense dashboard status and network interface configuration.



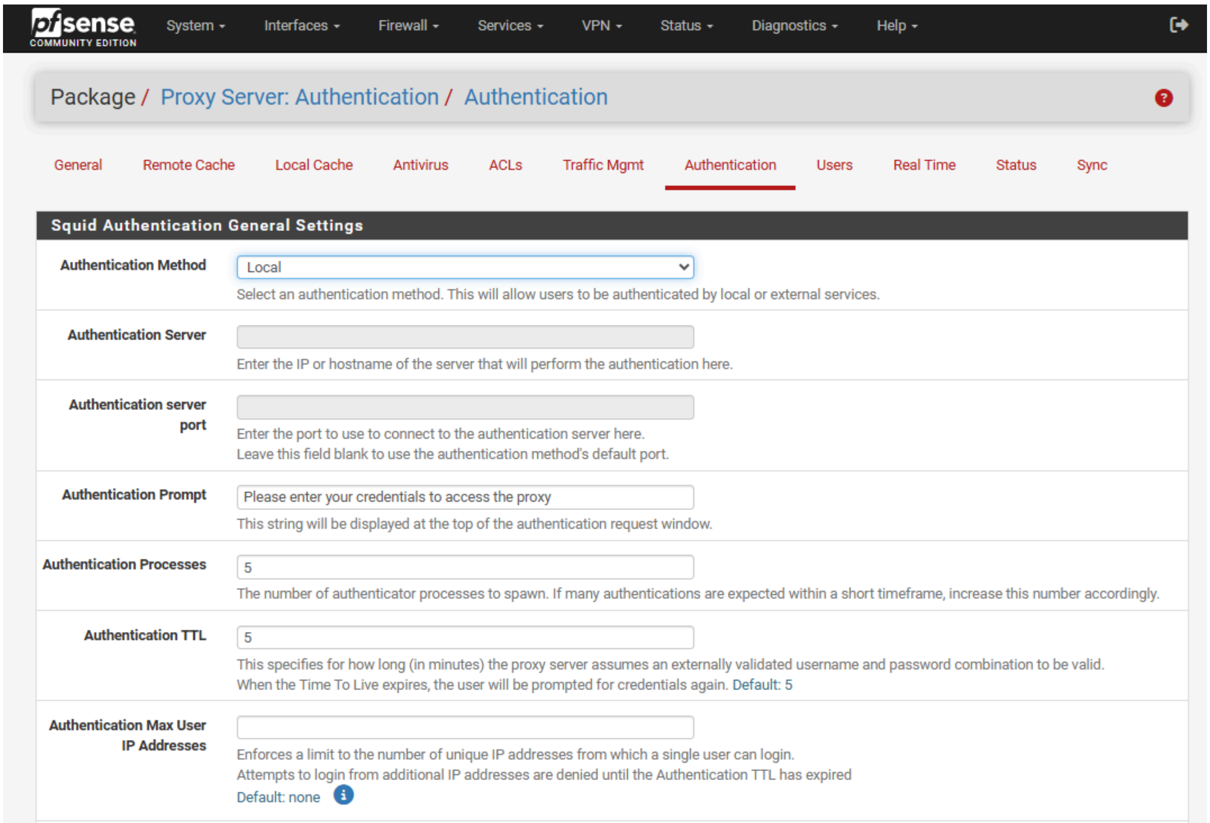
15) Search and install Squid, SquidGuard, and Lightsquid packages.



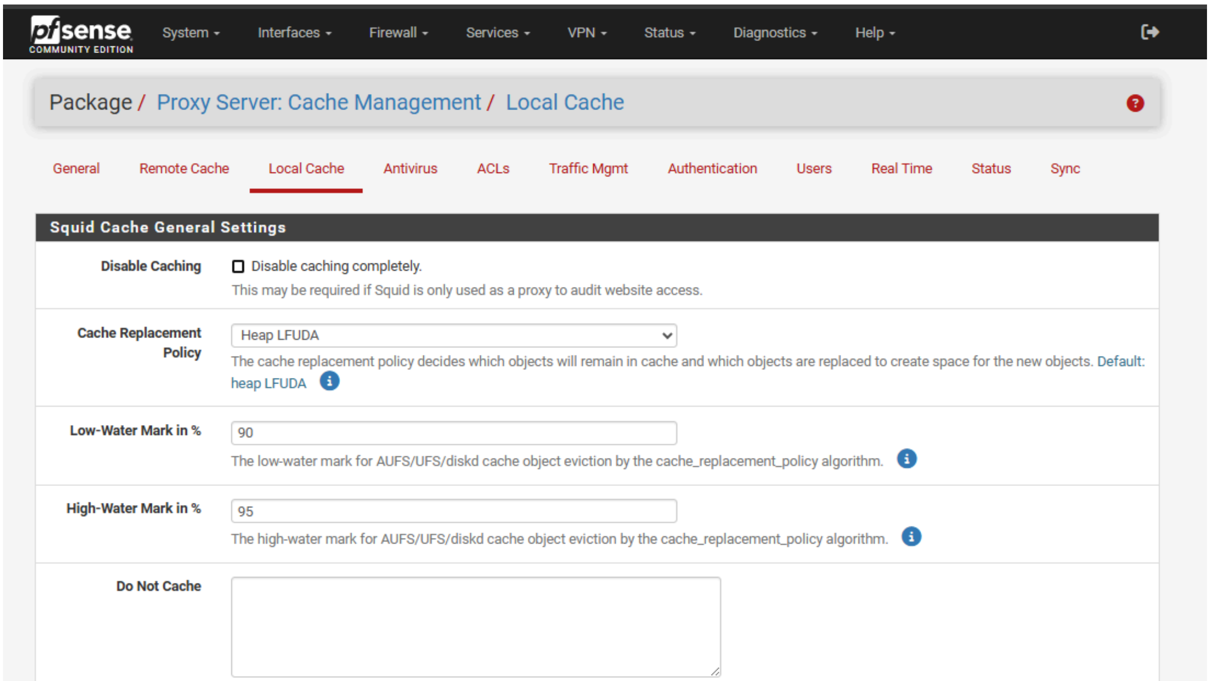
16) Verify successful installation of proxy-related packages.



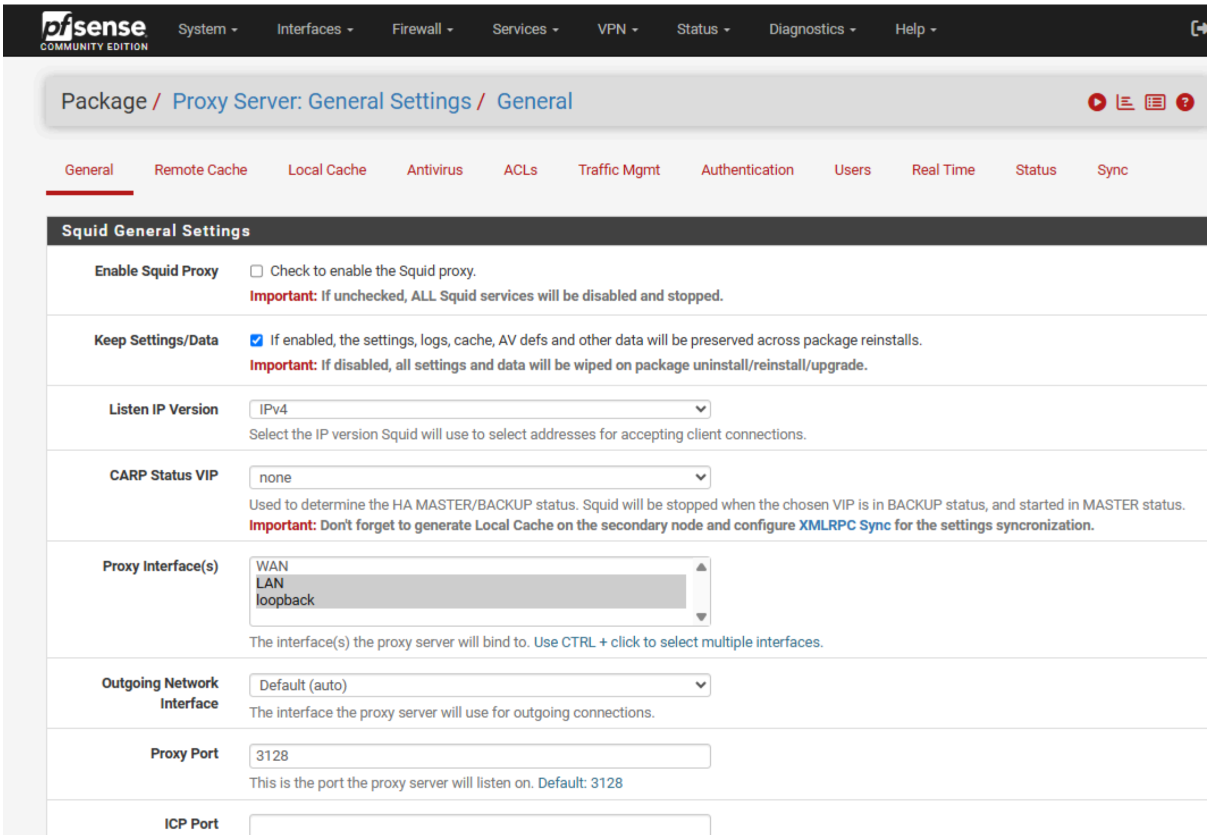
17) Configure Squid local user authentication settings.



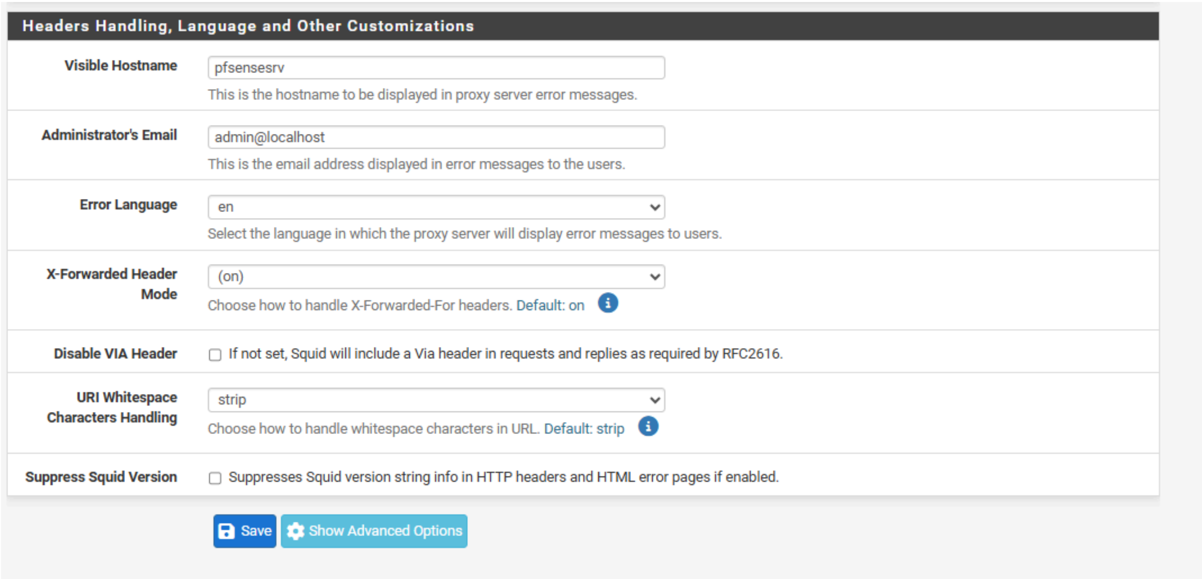
18) Configure Squid cache management and local cache parameters.



19) Enable Squid proxy service and select listening interfaces.



20) Configure proxy server hostname, email, language, and header options.

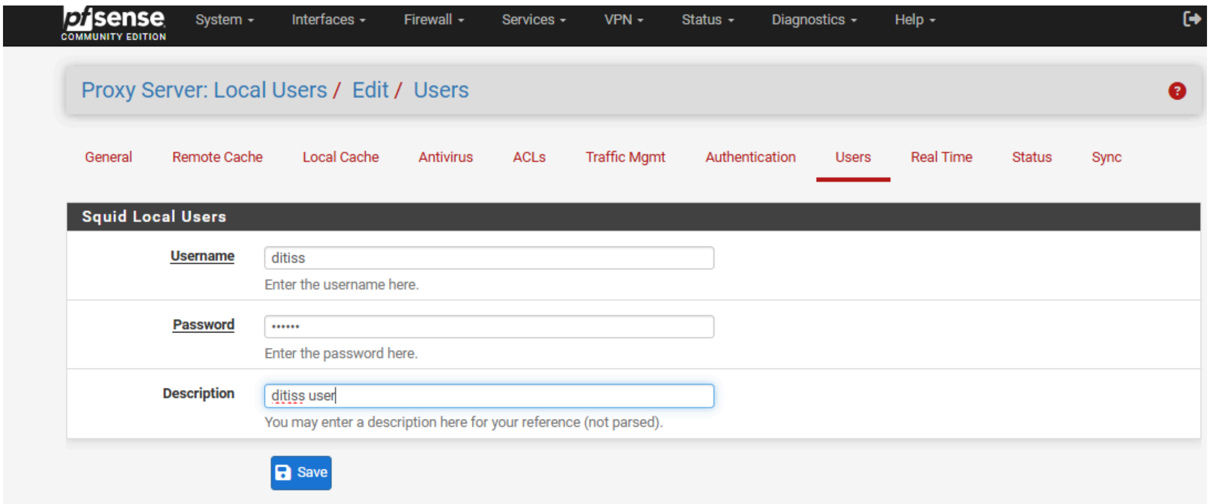


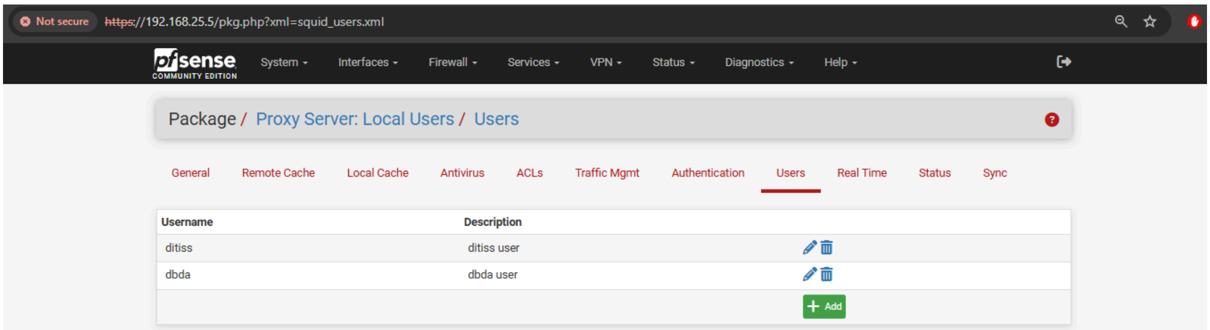
<u>Perform the following tasks:</u>

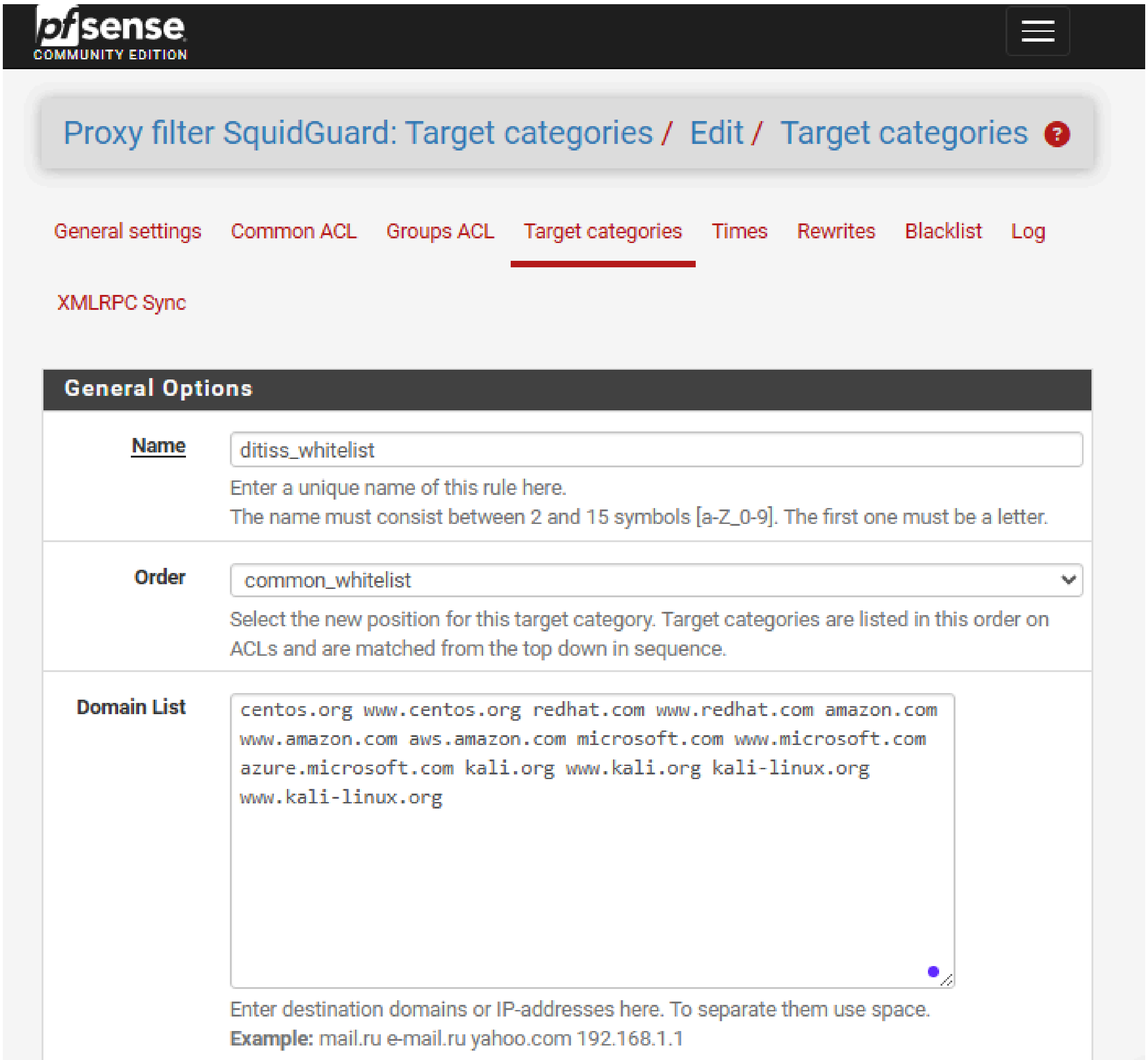
1. Create a user named <u>ditiss</u>

<u>Allow access only to:</u>

- CentOS website
- Red Hat website
- Amazon websites
- Microsoft Cloud websites
- Kali Linux website



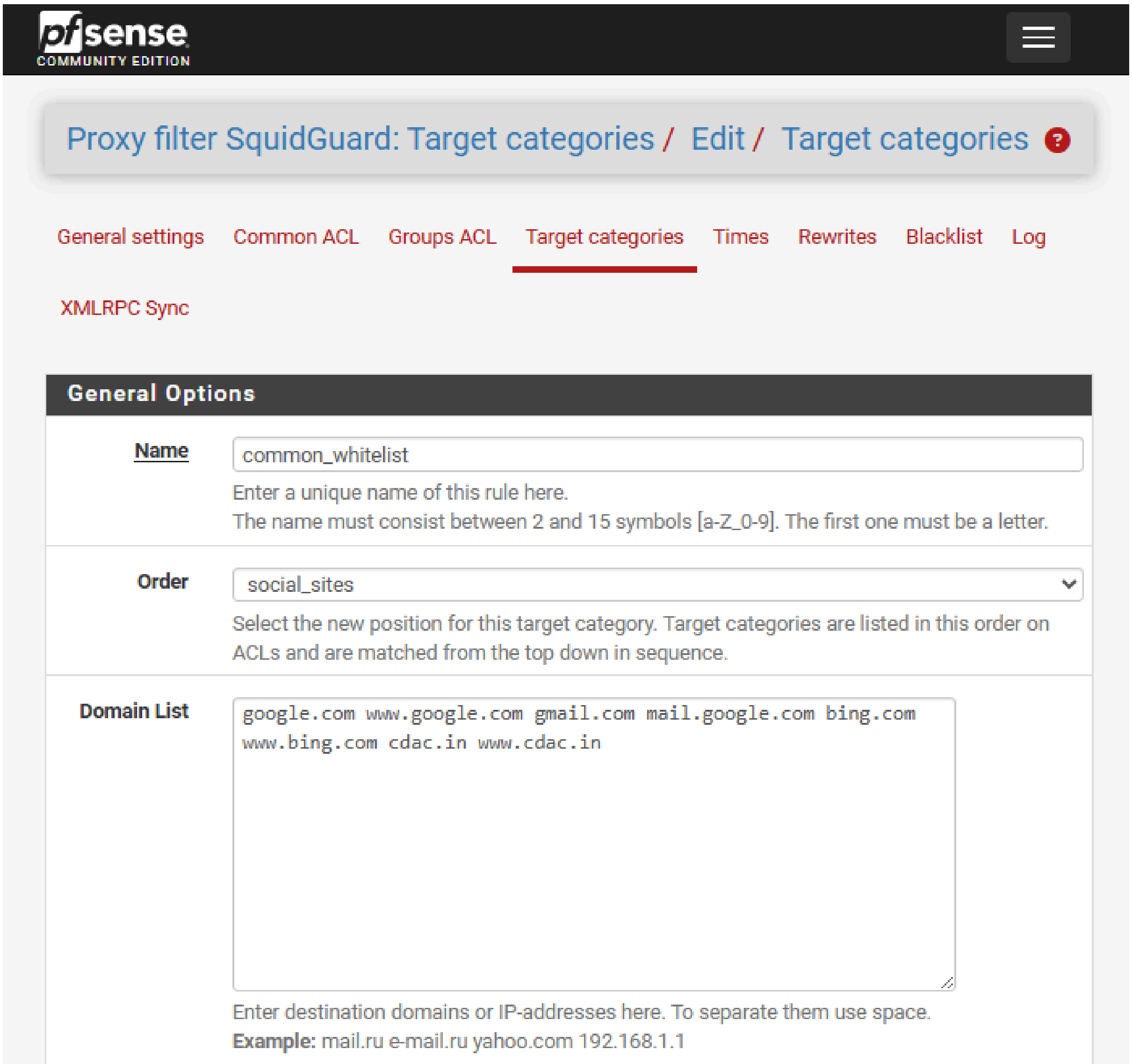

Not secure
https://192.168.25.5/pkg.php?xml=squid_users.xml
pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Package / Proxy Server: Local Users / Users
General
Remote Cache
Local Cache
Antivirus
ACLs
Traffic Mgmt
Authentication
Users
Real Time
Status
Sync
Username
Description
ditiss
ditiss user
dbda
dbda user
Add

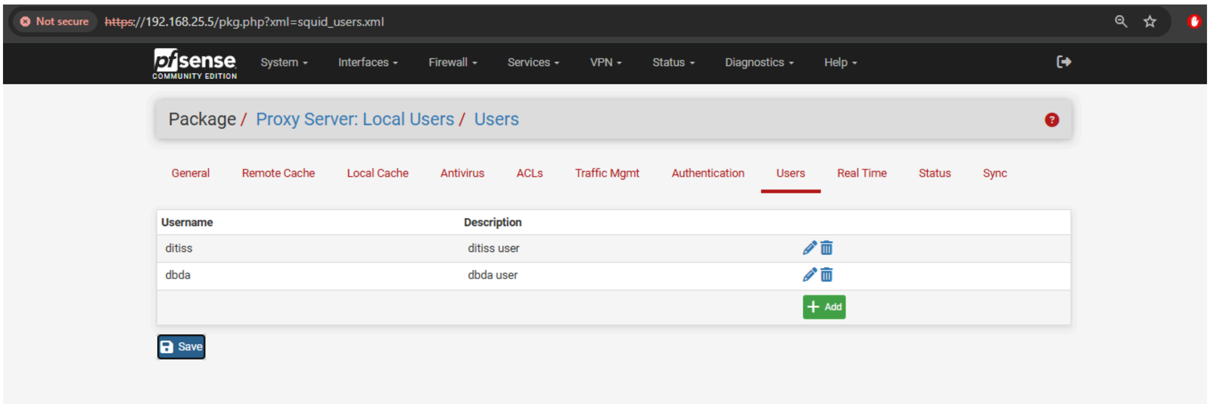

pfsense COMMUNITY EDITION
Proxy filter SquidGuard: Target categories / Edit / Target categories
General settings
Common ACL
Groups ACL
Target categories
Times
Rewrites
Blacklist
Log
XMLRPC Sync
General Options
Name
ditiss_whitelist
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.
Order
common_whitelist
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.
Domain List
centos.org www.centos.org redhat.com www.redhat.com amazon.com www.amazon.com aws.amazon.com microsoft.com www.microsoft.com azure.microsoft.com kali.org www.kali.org kali-linux.org www.kali-linux.org
Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

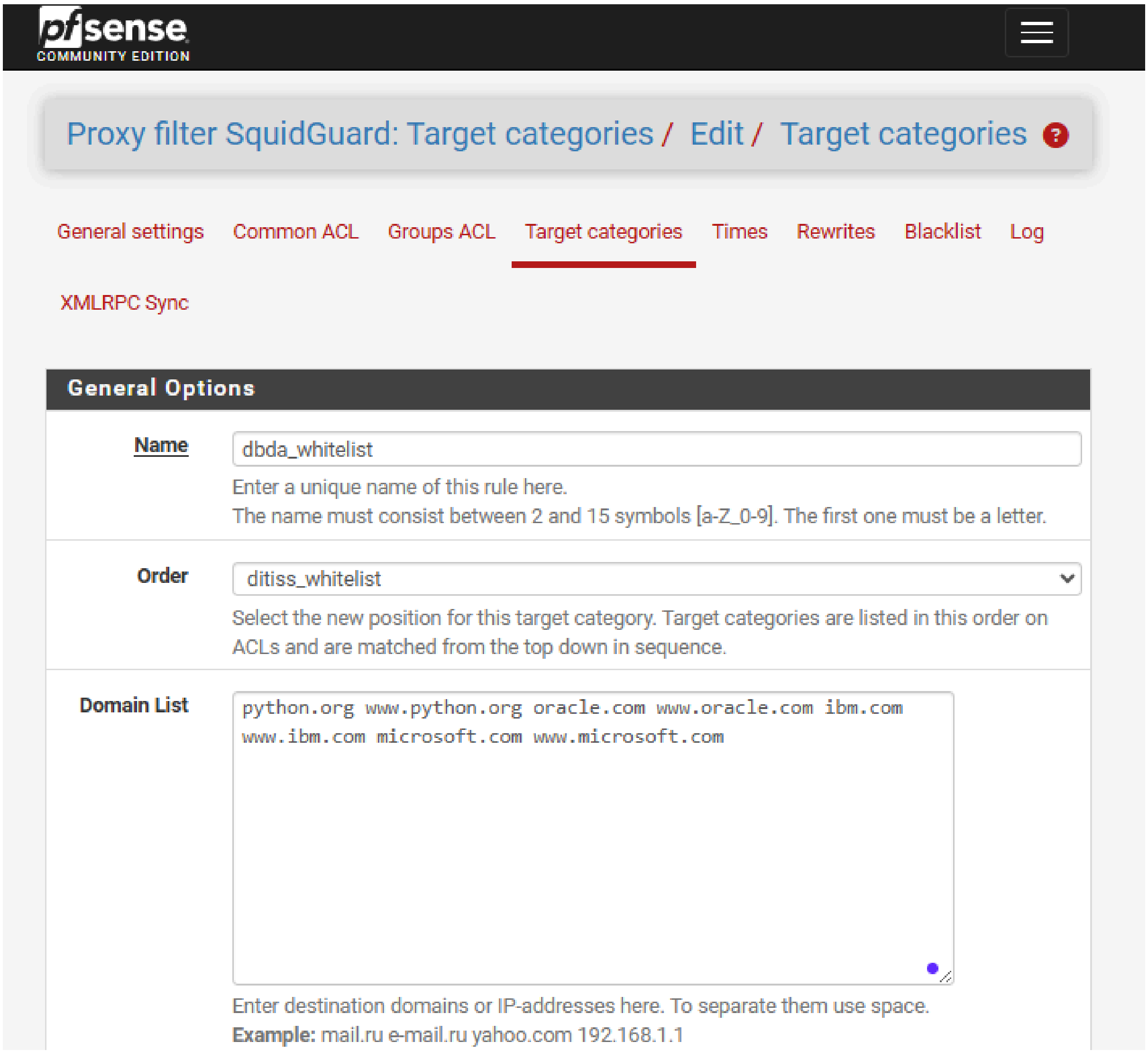
2. Create another user named <u>dbda</u>

<u>Allow access only to:</u>

- Python websites
- Oracle websites
- IBM websites
- Microsoft websites

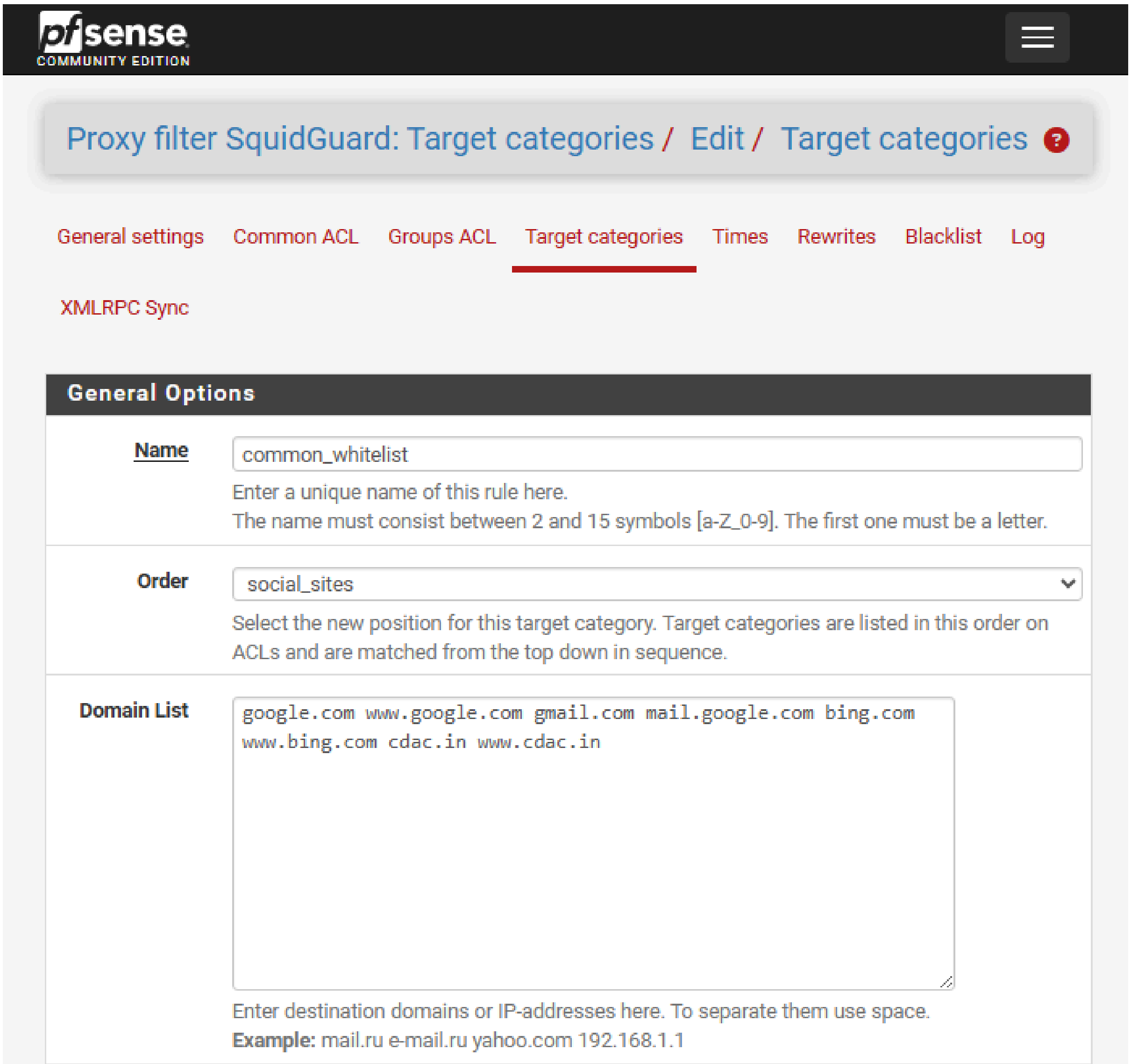


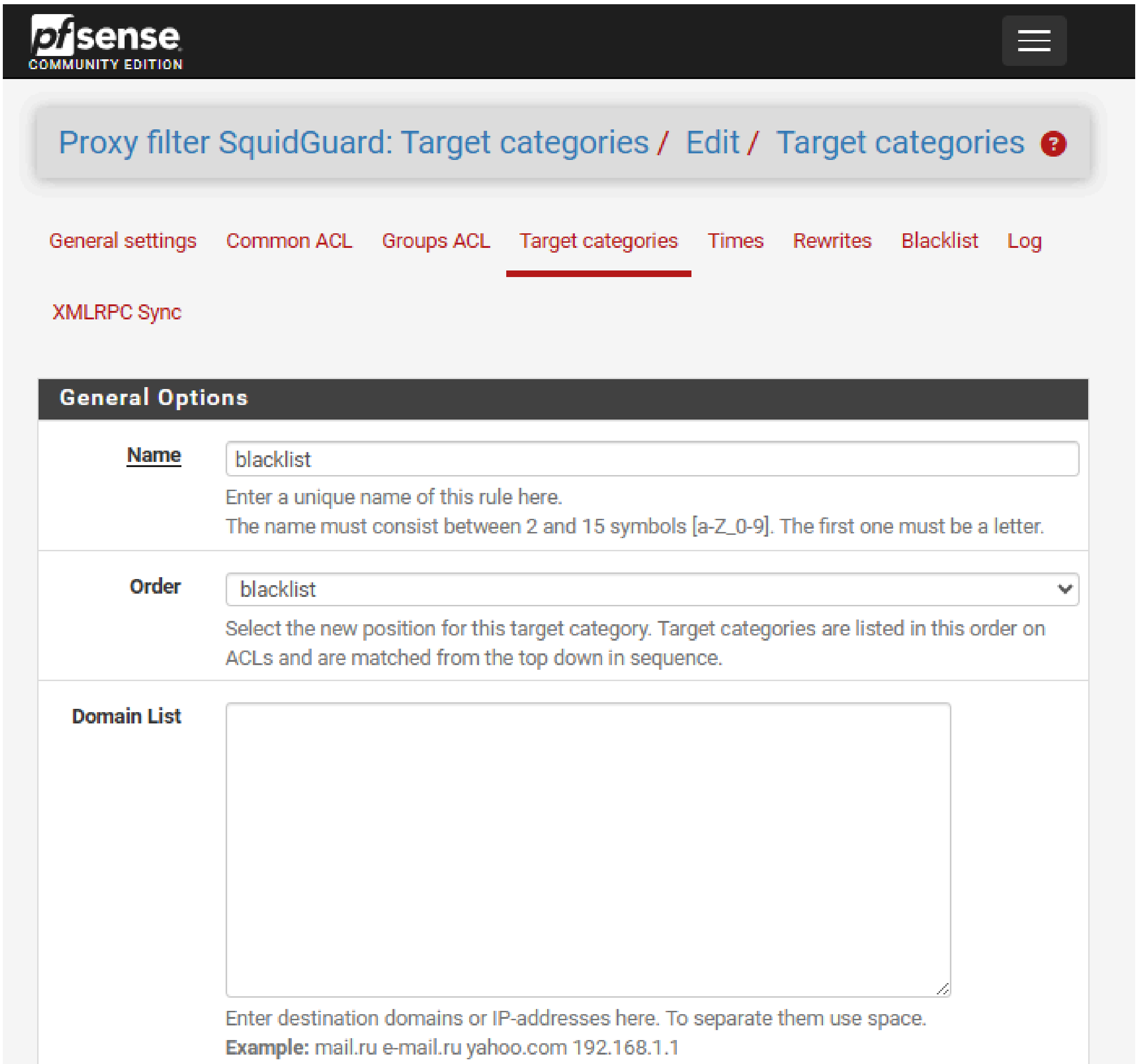

Not secure
https://192.168.25.5/pkg.php?xml=squid_users.xml
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
Package / Proxy Server: Local Users / Users
General
Remote Cache
Local Cache
Antivirus
ACLs
Traffic Mgmt
Authentication
Users
Real Time
Status
Sync
Username
Description
ditiss
ditiss user
dbda
dbda user
Add
Save


Proxy filter SquidGuard: Target categories / Edit / Target categories
General settings
Common ACL
Groups ACL
Target categories
Times
Rewrites
Blacklist
Log
XMLRPC Sync
General Options
Name
dbda_whitelist
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.
Order
ditiss_whitelist
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.
Domain List
python.org www.python.org oracle.com www.oracle.com ibm.com www.ibm.com microsoft.com www.microsoft.com
Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

3. For all users, allow access to:

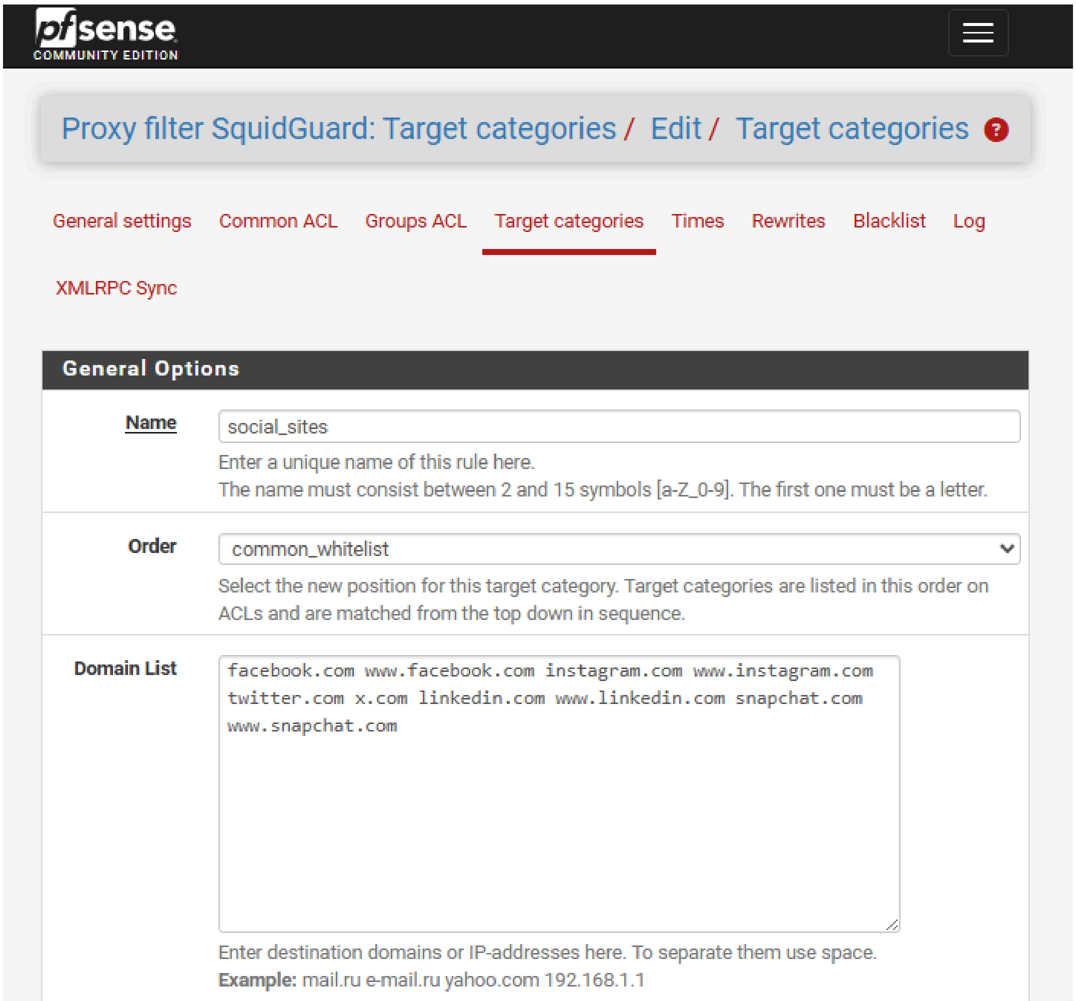
- Google websites
- CDAC websites
- Bing websites
- Gmail websites

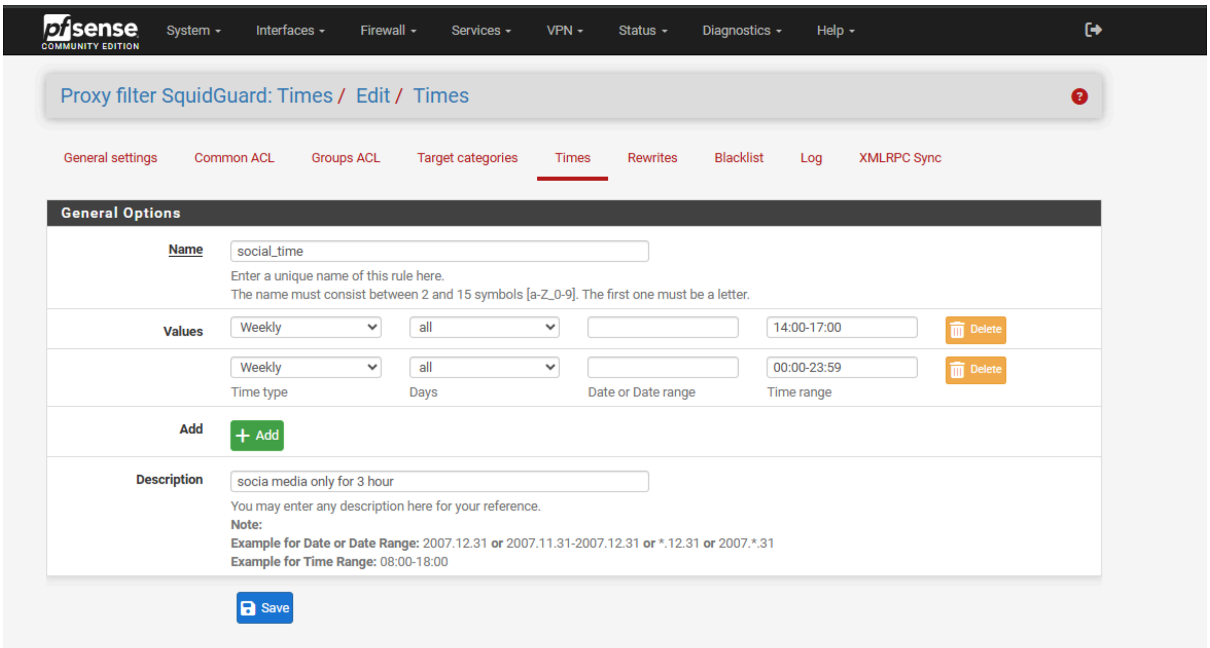


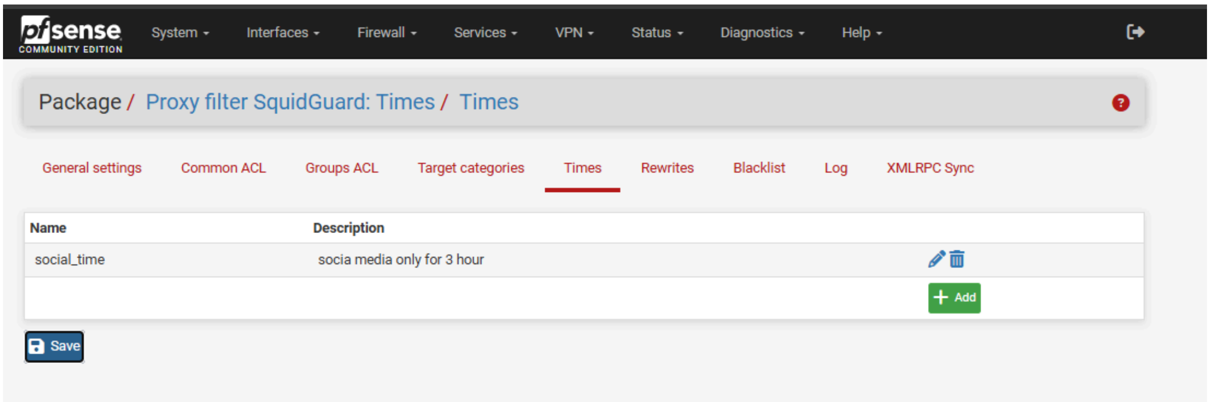


pfsense
COMMUNITY EDITION
Proxy filter SquidGuard: Target categories / Edit / Target categories
General settings
Common ACL
Groups ACL
Target categories
Times
Rewrites
Blacklist
Log
XMLRPC Sync
General Options
Name
blacklist
Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.
Order
blacklist
Select the new position for this target category. Target categories are listed in this order on ACLs and are matched from the top down in sequence.
Domain List
Enter destination domains or IP-addresses here. To separate them use space.
Example: mail.ru e-mail.ru yahoo.com 192.168.1.1

4. Allow access to social networking websites only for 3 hours (using a time-based access policy).







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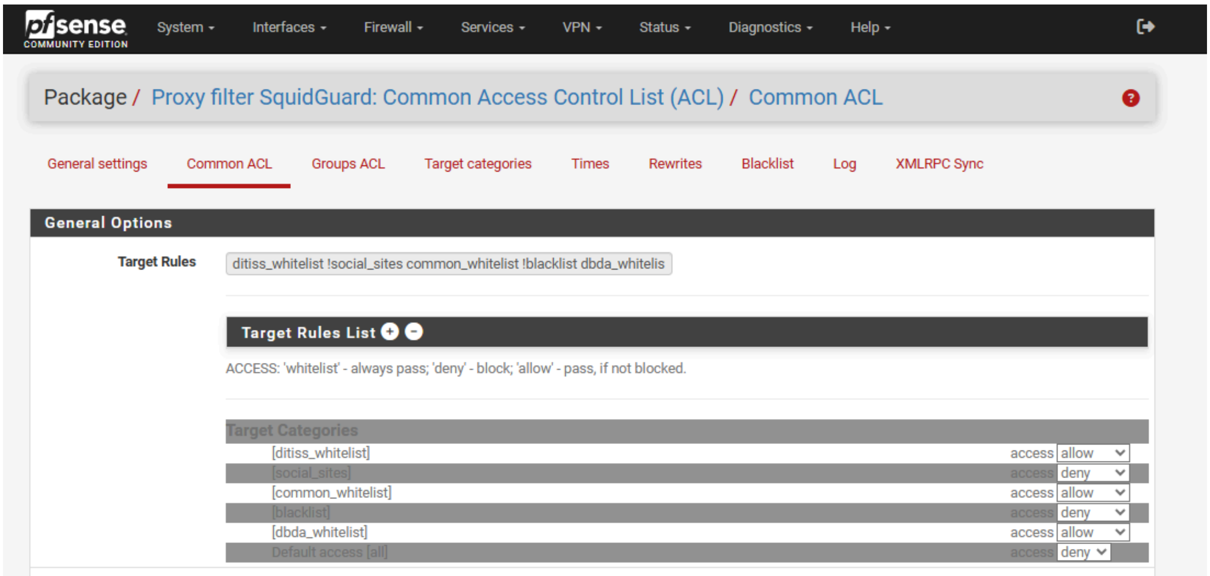
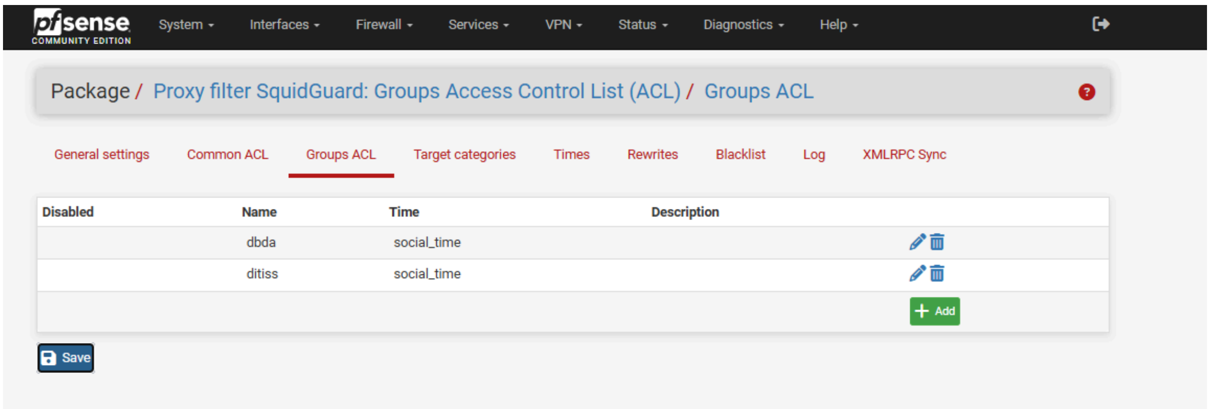
a) Configure Common ACL target rules in SquidGuard.



b) Create user groups in SquidGuard Groups ACL.



c) Create ACL rule for user 'ditiss' and assign whitelist permissions.

Name

ditiss

Enter a unique name of this rule here.
The name must consist between 2 and 15 symbols [a-Z_0-9]. The first one must be a letter.

Order

dbda

Select the new position for this ACL item. ACLs are evaluated on a first-match source basis.
Note:
Search for a suitable ACL by field 'source' will occur before the first match. If you want to define an exception for some sources (IP) from the IP range, put them on first of the list.
Example:
ACL with single (or short range) source ip 10.0.0.15 must be placed before ACL with more large ip range 10.0.0.0/24.

Client (source)

'ditiss'

Enter client's IP address or domain or "username" here. To separate them use space.
Example:
IP: 192.168.0.1 - **Subnet:** 192.168.0.0/24 or 192.168.1.0/255.255.255.0 - **IP-Range:** 192.168.1.1-192.168.1.10
Domain: foo.bar matches foo.bar or *.foo.bar
Username: 'user1'
Ldap search (Ldap filter must be enabled in General Settings):
ldapusersearch ldap://192.168.0.100/DC=domain,DC=com?sAMAccountName?sub?(&(sAMAccountName=%s)(memberOf=CN=it%2cCN=Users%2cDC=domain%2cDC=com))
Attention: these line don't have break line, all on one line

Time

social_time

Select the time in which 'Target Rules' will operate or leave 'none' for rules without time restriction. If this option is set then in off-time the second ruleset will operate.

Target Rules

ditiss_whitelist social_sites common_whitelist !blacklist !all [all]

Target Rules List

ACCESS: 'whitelist' - always pass; 'deny' - block; 'allow' - pass, if not blocked.

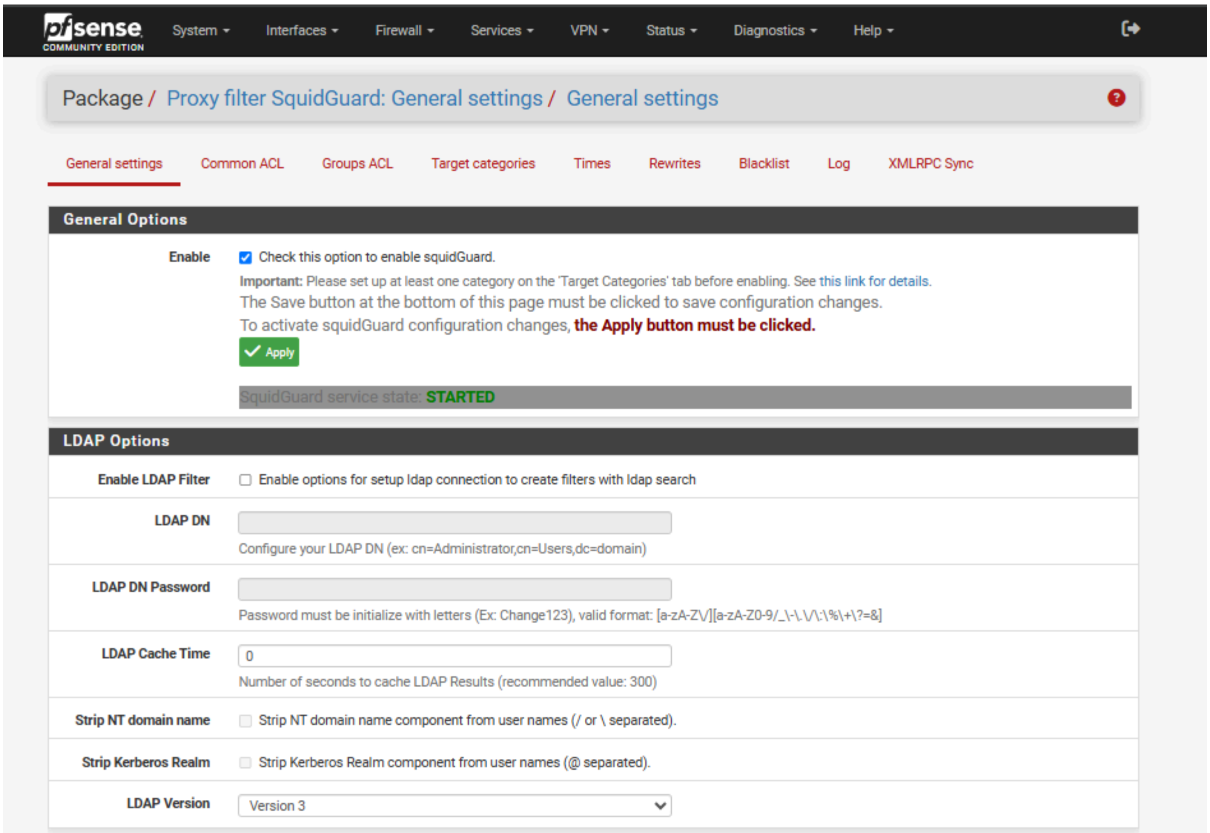
Target Categories			Target Categories for off-time (If 'Time' not defined, this is column will be ignored.)		
[ditiss_whitelist]	access	allow	[ditiss_whitelist]	access	---
[social_sites]	access	allow	[social_sites]	access	allow
[common_whitelist]	access	allow	[common_whitelist]	access	---
[blacklist]	access	deny	[blacklist]	access	---
[dbda_whitelist]	access	---	[dbda_whitelist]	access	---
Default access [all]	access	deny	Default access [all]	access	allow

Do not allow IP-Addresses in URL

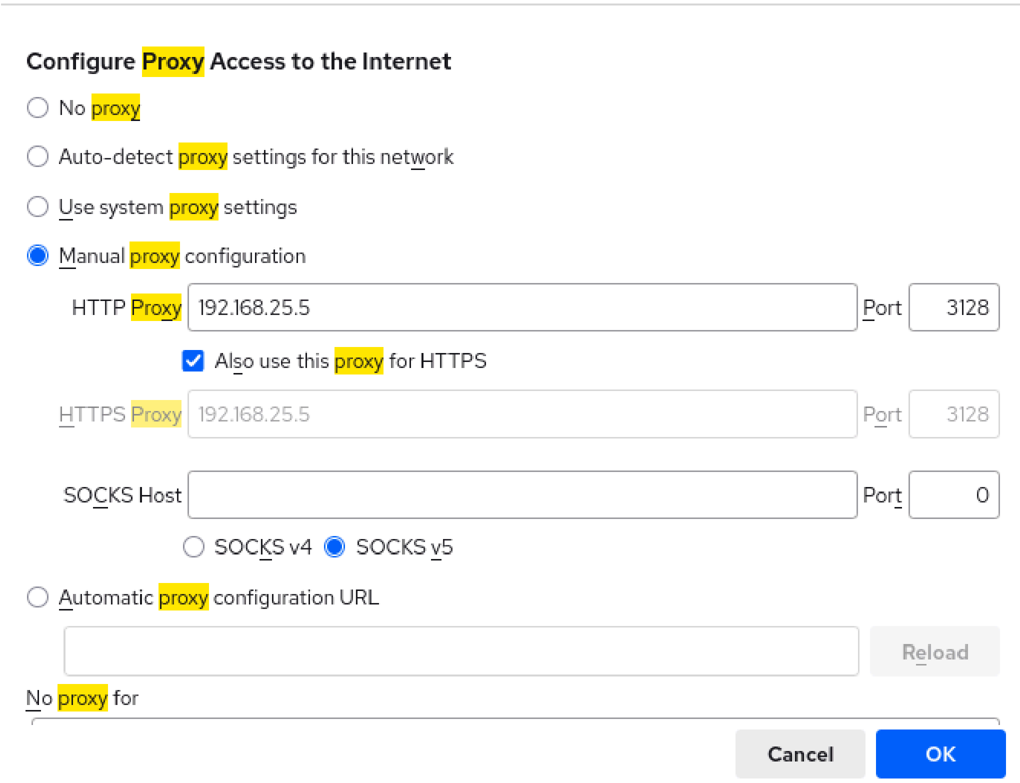
☐ To make sure that people do not bypass the URL filter by simply using the IP-Addresses instead of the FQDN you can check this option. This option has no effect on the whitelist.

d) Create ACL rule for user 'dbda' and define access permissions.

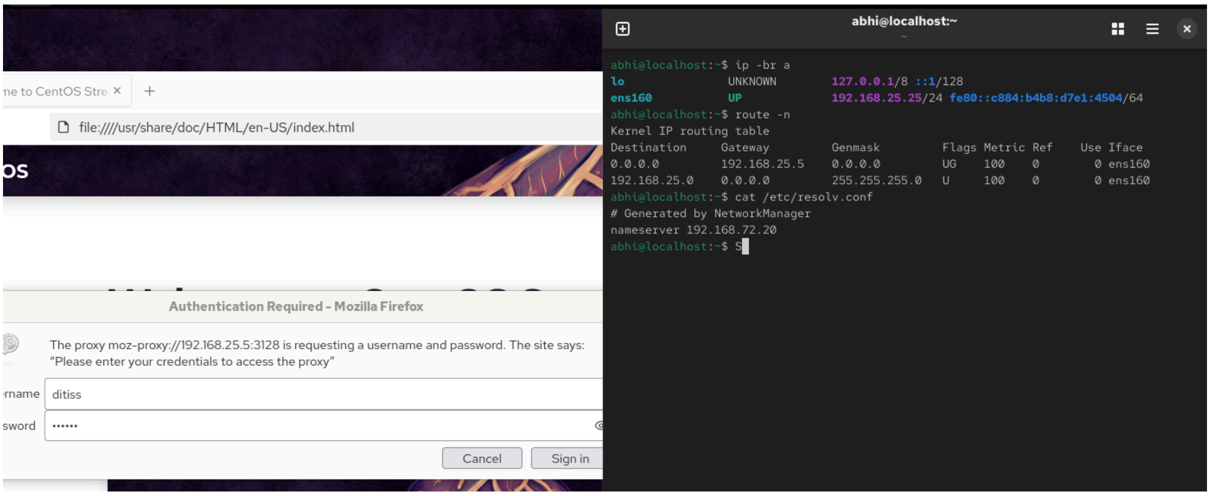
e) Enable and start SquidGuard URL filtering service.



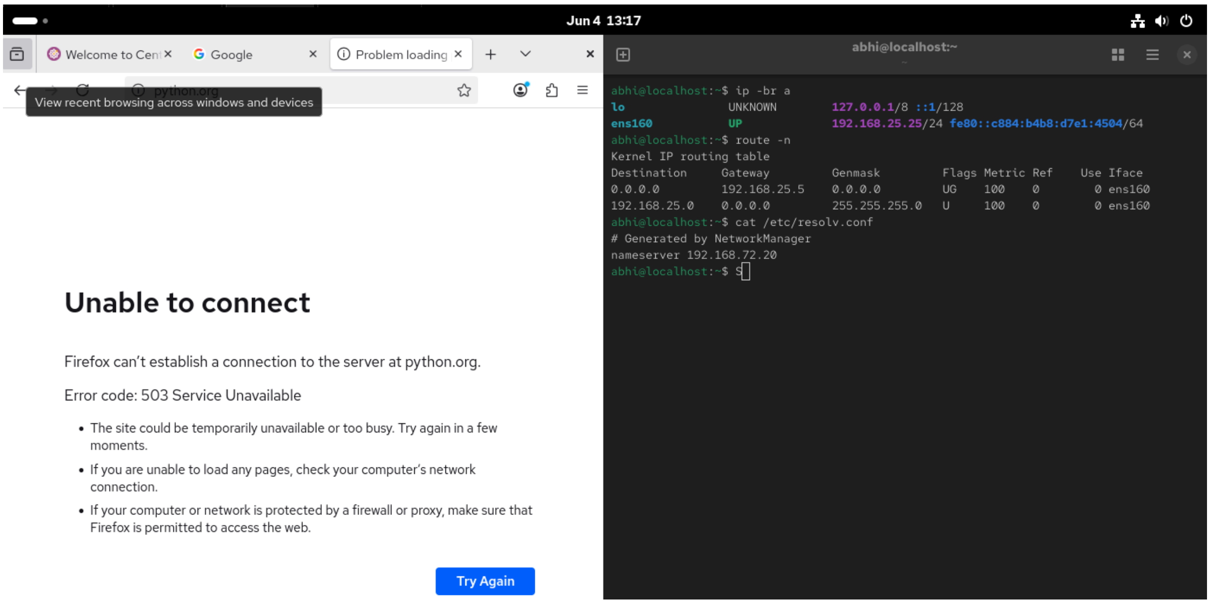
f) Configure client browser to use proxy server (192.168.25.5:3128).



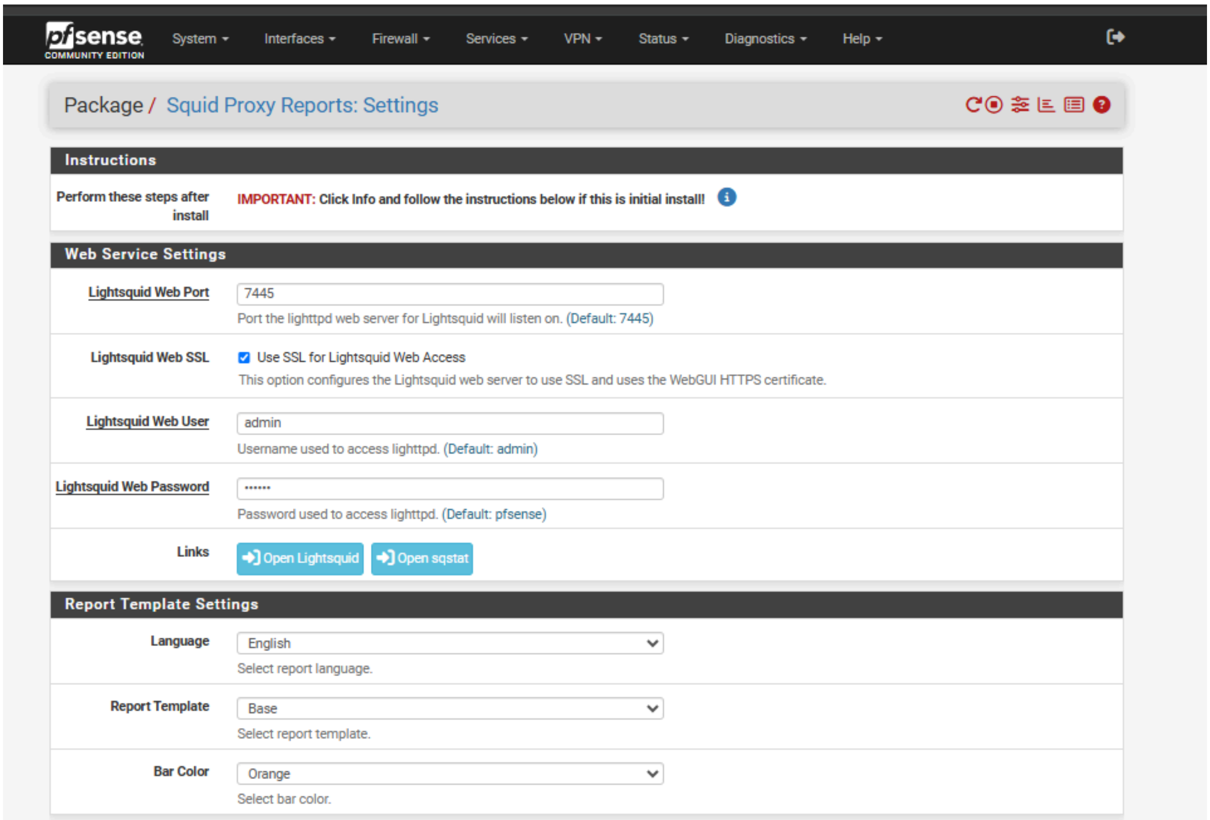
g) Authenticate to the proxy using user credentials.



h) Verify website access restrictions through SquidGuard filtering.



i) Configure Lightsquid reporting and generate proxy usage reports.



j) View and analyze generated user web access reports in Lightsquid.